

Isaac Mubiru

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PROFESSIONAL SUMMARY

Senior Backend Engineer with 6+ years building scalable TypeScript/Node.js systems and real-time data pipelines. Led zero-downtime migration of 20+ production applications from Parse/MongoDB to Firebase/GCP, designing event-driven architectures with Pub/Sub that mirror modern AI/ML inference pipelines. Deep experience with API design, cloud platforms (GCP, AWS), and database systems including MongoDB and Firestore. Proven ability to operate with high ownership and minimal supervision, delivering complex infrastructure projects while communicating cross-functionally with engineering and ops teams. Eager to apply data infrastructure expertise to building AI/ML-powered features at scale.

CORE COMPETENCIES

TypeScript / Node.js

React & Frontend Development

Scalable API Design

Real-time Data Pipelines

Cloud Platforms (GCP, AWS)

Database Systems (MongoDB, Firestore)

Event-Driven Architecture

Cross-Functional Collaboration

WORK EXPERIENCE

MTailor

September 2022 – Present

Software Engineer · US Remote

- **Led zero-downtime migration** of 20+ production applications from Parse/MongoDB to Firebase/GCP, reporting directly to CTO — architecture pattern applicable to AI/ML inference service deployments
- Designed and implemented **real-time two-way sync pipeline** between MongoDB and Firestore using Node.js and Google Pub/Sub for message processing — event-driven architecture used in modern ML feature stores
- Built **scalable APIs** connecting database capabilities with real-time data systems, handling concurrent production traffic during phased migration
- Developed 3D visualization feature using React frontend and video overlay with ffmpeg, **increasing buyer conversion**
- Implemented Express Shipping feature generating **\$40 additional revenue per order**
- Reduced infrastructure costs by **\$5,000/month** through AWS-to-GCP migration and architecture optimization

MTailor

March 2022 – July 2022

Contractor · US Remote

- Built **TypeScript migration scripts** to read data from MongoDB, run complex processing logic, and write to Firestore
- Created technical documentation for fellow engineers on new Firebase SDKs (Node.js and Python)
- Trained operations team on Firebase Dashboard, backup procedures, and **monitoring observability**

Dr Wealth

August 2021 – January 2022

Software Engineer · Singapore Remote

- Built **data-oriented interfaces** querying 2M+ Firestore records for real-time stock price visualization
- Extended **backend APIs** using Firebase Cloud Functions and Express.js hosted on Heroku
- Developed responsive UI pages using React, Tailwind, and HTML for real-time financial data display

Mind2matter.co

January 2022 – July 2022

Software Engineer · US Remote

- Built **full-stack applications** with Node.js backends and React frontends under tight timelines
- Delivered DeFi application backends using Web3 and TypeScript/Node.js

CodeBits

January 2020 – July 2021

Software Engineer · Uganda Remote

- **Led team of 4 developers** building legal tech systems for NGOs, demonstrating cross-functional communication
- Architected **event-driven microservices** backend using Apache Kafka, Docker, and Kubernetes — patterns used in ML inference pipelines
- Built USSD service communicating with Node.js backend via gRPC for **scalable API design**

- Developed cross-platform mobile app using React Native with Firebase push notifications

EDUCATION

Bachelor of Science in Computer Engineering — [Makerere University](#)
Kampala, Uganda

2018 – 2022

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, Go (learning) **Backend:** Node.js, Express.js, Firebase Cloud Functions, gRPC
Frontend: React, React Native, HTML, CSS, Tailwind **Cloud:** GCP (Pub/Sub, Cloud Functions, GCS), AWS (S3, EC2, EBS)
Databases: MongoDB, Firestore, SQL **Infrastructure:** Docker, Kubernetes, Apache Kafka **Tools:** Git, Linux, ffmpeg